

A042 · Understand the Effect of Genetic Variants on Alzheimer's Disease

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Alzheimer's disease (AD) is a neurodegenerative disease with a strong genetic component. Genome-wide association studies (GWAS) have identified thousands of genetic variants associated with clinical traits, including AD status and AD-related clinical traits. The mechanisms through which genetic variants exert their effects are not always clear, but the Genotype-Tissue Expression (GTEx) project has demonstrated that genetic variation can affect gene expression across human tissues, including the brain. However, GTEx contains a limited set of brain tissues, missing some brain regions that may be important for disease development. Our recent model, BRONTE, aims to overcome this deficit by projecting the effects of genetic variants across 103 brain tissues. We hypothesize that these results will help explain the biological mechanisms connecting genetic variation, gene expression, and Alzheimer's disease clinical traits. Using data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), we performed association tests between ~100 clinical traits, including PET imaging measures and biomarkers, and AD status, and identified p-tau levels and left hippocampal volume as traits associated with disease status. We then performed GWAS on p-tau levels and left hippocampal volume to identify genetic variants associated with those traits. For the associated variants, we investigate their associations with gene expression in specific brain regions using results from BRONTE. By integrating ADNI clinical and genetic data with GTEx-based gene expression imputation, this project provides a framework for understanding how genetic variation may influence Alzheimer's disease through changes in gene expression.